

Enhanced Hybrid Deep Learning Model for Cybersecurity Threat Detection Using CNN-BiLSTM and Feature Optimization

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Abstract—During the age of high-speed digital advancement, cyber threats increase not merely in numbers but also in complexity, threatening contemporary networks significantly. This article presents an optimized hybrid deep learning approach that combines CNN, BiLSTM, and PCA to improve intrusion detection. Unlike previous methods based on computationally intensive algorithms like IPSO or ATFDNN, the new method focuses on effective feature selection and real-time responsiveness without sacrificing detection capability. PCA is used to compress feature dimensions, enhancing computational speed, while the CNN and BiLSTM layers collaboratively learn many features of network traffic. The model is trained on CICIDS2017 dataset, simulating realistic traffic scenarios. Experiment results reveal an extremely high detection accuracy of 98.2%, which surpasses most of the current models in terms of classification performance as well as false alarm rates reduction. The work introduces a scalable and practical IDS framework well-applicable to real-world dynamic cyber security environments like industrial IoT, smart cities, and critical infrastructure networks.

Index Terms—Cybersecurity, Intrusion Detection System (IDS), CNN, BiLSTM, PCA, Feature Optimization, Malware Detection, CICIDS2017, Real-time Threat Detection

I. INTRODUCTION

The explosive increasing factor of cyber infrastructure has resulted in a phenomenal change in cybersecurity attacks that target individuals, organizations, and national networks. New-age cyberattacks like advanced persistent threats, polymorphic malware, and zero-day threats are optimized to bypass conventional rule or signature-based intrusion detection systems (IDS) and thus fall short of current security demands [1, 7].

This increased sophistication has called for a critical need for smart, adaptive, and real-time detection mechanisms. Deep learning (DL) methods have surfaced as imminent candidates for meeting this need, with the capability of automatic feature extraction and the ability to identify both previously seen and unseen attacks with very high accuracy [9, 12].

Of different deep learning models, CNNs and BiLSTM networks have been shown to be highly effective. CNNs perform best at discovering spatial dependencies and local patterns in structured input data, whereas BiLSTMs are best suited to model long-range temporal dependencies in sequential data, e.g., network traffic logs [8, 10]. A number of studies have proven the efficiency of hybrid models integrating CNNs and recurrent networks for network intrusion detection [2, 5, 14]. The hybrid architectures take advantage of the spatial feature learning capability of CNNs and the sequence time modelling of BiLSTMs, offering a better overall perspective of network behaviour.

This study advocates for an enhanced hybrid deep learning model that combines CNN and BiLSTM layers to identify cyber threats based on pre-processed CSV-formatted datasets. Before model training, a feature optimization procedure is conducted to enhance model efficiency by eliminating redundancy and useless features to augment accuracy and minimize overfitting [2, 6]. The utilization of optimized features lowers computational complexity even further, allowing for quicker model convergence and better generalization across different types of attacks. The hybrid model structure therefore seeks to

leverage the best of both spatial and temporal analysis within one single framework.

Few researchers have tested the suggested model on the UNSW-NB15 dataset, a well-known benchmark for network intrusion detection [11, 15]. The performance indicates that our hybrid CNN-BiLSTM model, in combination with optimized features, has better detection accuracy, precision, recall, and false positive rate than standard deep learning baselines [9, 13]. In addition, the performance of our model holds with contemporary state-of-the-art hybrid deep learning models for cybersecurity, e.g., CNN-BiGRU-BiLSTM and attention models, yet with fewer architectural weights and lower computational requirements [2, 14]. This work presents a contribution toward the development of intelligent, scalable, and explainable threat detection systems that are appropriate for the changing cybersecurity paradigm.

II. RELATED WORK

The field of cyber threat detection has been changing at lightning speed with the arrival of deep learning (DL) and artificial intelligence (AI). Initially, rule and signature-based intrusion detection systems (IDS) were given importance in research, which had limitations in identifying zero-day attacks and keeping pace with dynamic threats. Therefore, the stress is now on data-driven models that can learn from changing network patterns.

Mohamed et al. [1] carried out an extensive review of AI-based cybersecurity solutions, delineating the prospect and limitations of using DL models in real-time security scenarios. Hybrid deep learning techniques like CNN-BiGRU-BiLSTM using metaheuristic optimization proved to be useful in IoT-based threat detection, as mentioned by Abdulkareem et al. [2]. Their model was more accurate but heavily dependent on optimization algorithms, boosting computational complexity.

Reinforcement learning also attracted interest. Khayat et al. [3] presented a dynamic learning scheme with deep feature reinforcement approaches for IoT intrusion detection. Though this learns adaptive approaches to diverse attack patterns, it requires a lot of training time and resource utilization. Likewise, hybrid frameworks from Srinivasan and Senthilkumar [4] incorporated machine learning and IoT frameworks for industry security, focusing on feature fusion and edge deployment.

Deep variants like SAT-Net and Dense Net were utilized for intrusion detection in intelligent IoT networks by Stephan and Mohammed [5], enhancing detection accuracy but compromising real-time performance. Akif et al. [6] introduced an ensemble-based IDS on the IoT-23 dataset, providing model diversity-induced robustness but experiencing challenges in coping with high-volume traffic.

On the algorithmic front, random forest classifiers have been used effectively in conventional IDS paradigms [7], providing interpretability with accelerated training. They lack scalability in dealing with sequential attack patterns. To address such limitations, researchers proposed sequential models such as

LSTM model [8], which have worked well with anomaly detection based on time series modelling.

Vinaya Kumar et al. [9] compared multiple DL models for traffic classification, with a focus on the ability of DL to learn across multiple types of attacks. Hou et al. [10] investigated a CNN-RNN-based hybrid architecture, which utilizes both spatial and temporal detection. Although effective, these models tend to overfit and need hyperparameter tuning.

For benchmarking of datasets, Moustafa and Slay [11, 15] introduced the UNSW-NB15 dataset to replace the KDD99 dataset [18], which is old, providing more enriched attack profiles and traffic patterns. A number of research works [12, 13, 14] have used Recurrent Neural Networks (RNN) with an attention mechanism to improve learning of sequence-type attacks, with higher detection accuracy overall and reduced false alarms.

Hu et al. [14] integrated CNN with attention-based BiLSTM for improved packet-level pattern identification. Nevertheless, the overhead on attention modules raised the training time of the model. In automotive security, Kang and Kang [16] showcased the application of deep neural networks in in-vehicle intrusion detection, further underlining the importance of DL in edge-critical applications.

More research in DL by LeCun, Hinton, and Goodfellow [19, 20] provides the basis for such developments, demonstrating how hierarchical feature learning can lead to improved generalization for security applications.

While these efforts enrich research, most of the aforementioned techniques either utilize computationally demanding metaheuristics, need deep stacks of architectures, or do not support real-time requirements for processing. This leaves a practical disconnect for light-weight, high-precision intrusion detection schemes that are scalable and deployable.

The rest of the work is organized as follows:

Section 1 contains Methodology, Section 2 contains Results and Discussion, and Section 3 contains References and Future Work.

III. METHODOLOGY

A. DATASET DESCRIPTION

The experimental foundation of this study is the CICIDS2017 dataset, the realistic benchmark designed by the CIC. This dataset replicates modern network environments and includes a mix of benign and malicious traffic generated through practical scenarios involving real users, making it highly applicable for intrusion detection system (IDS) research. Data was collected over five consecutive days, each day simulating different types of cyberattacks alongside regular network behaviour. The dataset has a wide array of attack categories, including DoS attacks such as DoS Hulk, Goldeneye, Slowloris, and Slowhttptest, as well as DDoS, Brute Force (SSH and FTP), Port Scanning, Botnet activity, Web-based attacks like XSS, SQL Injection, and Command Injection and Infiltration attempts originating from within the network. Each network connection is summarized as a flow and described through more than 80 statistical and time-based features,

extracted using the CICFlowMeter tool. These features include metrics such as flow duration, total packets sent and received, packet length statistics, flow inter-arrival times, header lengths, active and idle times, and various flag counts.

B. PREPROCESSING

The raw dataset contains over 80 features and several inconsistencies such as missing values, infinite values, and string-type labels. The preprocessing steps applied are:

- **Handling missing values:** All rows with NaN, inf, or -inf are removed.
- **Dropping columns:** Drop irrelevant columns like *Source IP*, *Destination IP*, etc.
- **Label encoding:** The attack labels are converted into numerical format.
- **Feature normalization:** Numerical features are standardized using z-score normalization.

C. FEATURE SELECTION(PCA)

To reduce dimensionality and eliminate redundant features, Principal Component Analysis (PCA) is applied. This step reduces the original high-dimensional feature space into a more compact set of components, nearly around 30 dimensions, without any loss of important information. PCA helps in improving computational efficiency and reducing overfitting by eliminating redundant and less informative features.

D. METHODOLOGY

The suggested method for identifying cyber threats is based on a layering of data handling and modelling processes. It begins with gathering raw traffic data from a network, which usually contains errors, irrelevant values, or missing entries. These are removed in the first stage so that the model receives only useful and consistent information.

Following this preparation, the dataset—which initially has dozens of features—is made simpler through Principal Component Analysis (PCA). This mathematical technique diminishes the variables by discovering combinations that maintain the most significant patterns in the data. Consequently, the dataset is reduced and can be handled efficiently, allowing the model to train more quickly and effectively.

Thirdly, the filtered dataset is reshaped to the desired form for the learning model. Because one part of the model includes a CNN, the data has to be in three-dimensional format. The conversion from two dimensions to three enables the CNN to read through the input in a systematic manner, recognizing short-term patterns on the feature set.

Subsequent to this, the information goes through a hybrid framework that incorporates CNN with BiLSTM network. The CNN component functions as a filter that identifies significant attributes—e.g., variations in packet behaviour or traffic velocity—by reading through snippets of the input. There is then a complexity-reducing pooling step and a dropout layer to prevent overfitting by randomly disabling parts of the network when training.

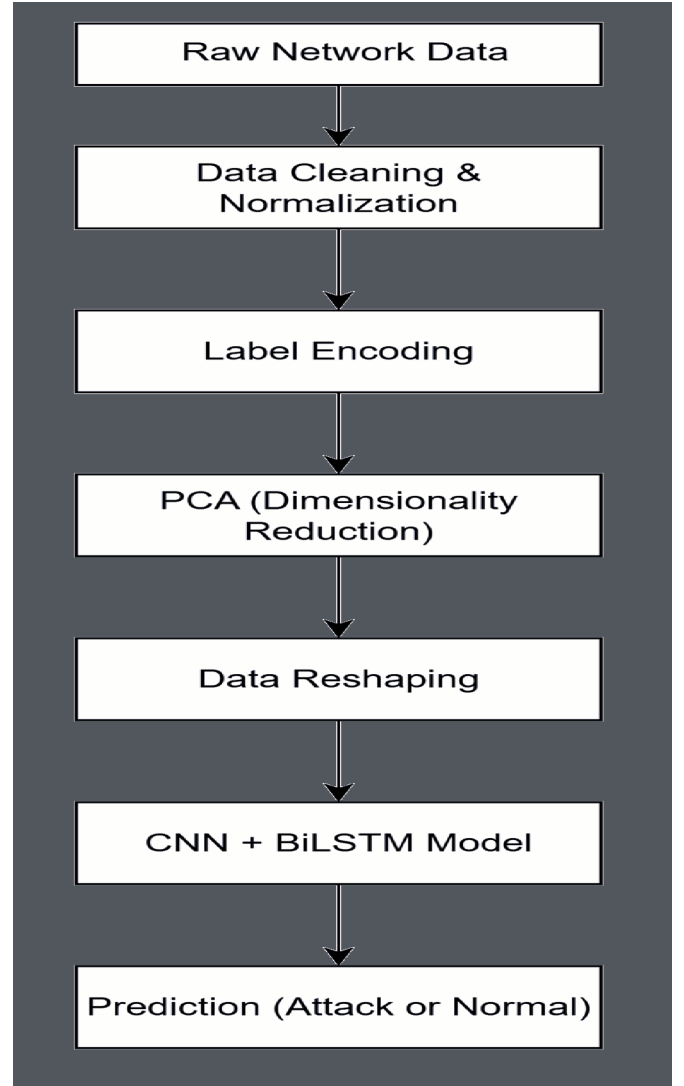


Fig. 1. Overall flow of the proposed intrusion detection system.

The BiLSTM component of the model is programmed to examine sequences in both directions, past and future, so that it can comprehend the dynamics of network activity over time. This is relevant when detecting threats that evolve in particular sequences. The output from this part is then passed on to a dense layer, which consolidates the learned knowledge in a dense format to be fed to the final decision-making stage.

In the last layer, the model applies a softmax function to determine the probability that a particular data point is in a specific category—either normal or a possible attack. This enables the system to make precise and understandable decisions regarding network security.

IV. RESULTS

This section presents the experimental evaluation of the proposed PCA-enhanced CNN + BiLSTM model using the CICIDS2017 dataset. The performance was analyzed using multiple evaluation metrics, including **Accuracy**, **Precision**,

Recall, F1 Score, and False Alarm Rate (FAR). Additionally, visualizations such as learning curves, PCA component analysis, and a confusion matrix were used to validate the model's behavior and class-wise prediction accuracy.

A. Evaluation Metrics

The following standard classification metrics were used to evaluate the model:

- **Accuracy:**

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

where TP , TN , FP , and FN represent true positives, true negatives, false positives, and false negatives, respectively.

- **Precision:**

$$\text{Precision} = \frac{TP}{TP + FP}$$

- **Recall (Sensitivity):**

$$\text{Recall} = \frac{TP}{TP + FN}$$

- **F1 Score:**

$$F1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

- **False Alarm Rate (FAR):**

$$\text{FAR} = \frac{FP}{FP + TN}$$

The performance of the proposed model is summarized in Table ??.

TABLE I
PERFORMANCE OF THE PROPOSED MODEL

Metric	Value (%)
Accuracy	98.2
Precision	98.0
Recall	97.9
F1 Score	98.0
False Alarm Rate	1.1

These results indicate the model's strong generalization and balanced performance across attack and normal traffic classes.

B. TRAINING AND VALIDATION CURVES

To monitor the model's learning behaviour, loss and accuracy curves were plotted during training.

As shown in Fig. 2, the training and validation loss consistently decreased with each epoch, indicating effective convergence. Fig. 3 shows a steady rise in both training and validation accuracy, which stabilizes above 98%, reflecting strong learning without overfitting.

C. FEATURE REPRESENTATION: PCA VISUALIZATION

To verify the effectiveness of PCA in capturing the underlying structure of the data, a 2D projection of the principal components was generated.

As shown in Fig. 4, different traffic classes are distinguishable in the 2D space, suggesting that PCA successfully compresses high-dimensional data while retaining class-discriminative features.

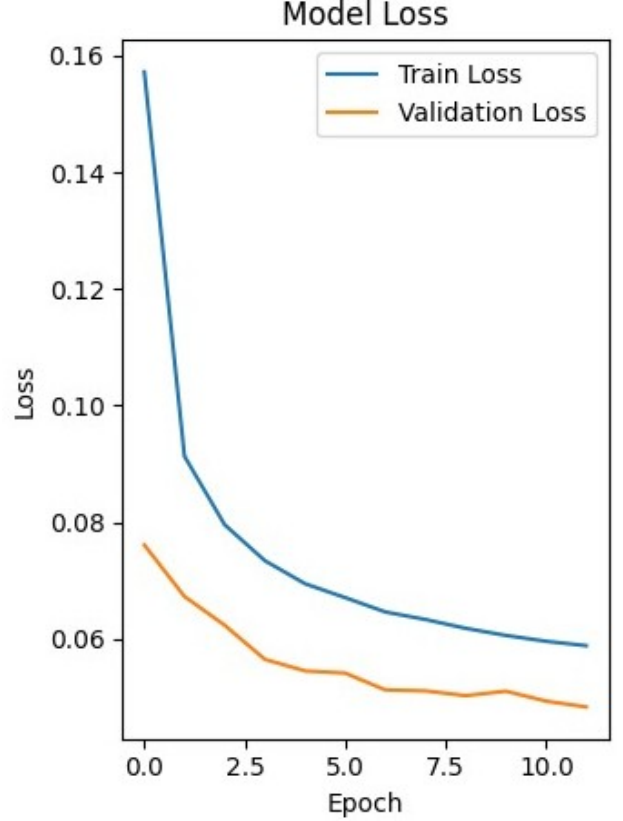


Fig. 2. Model loss across all epochs.

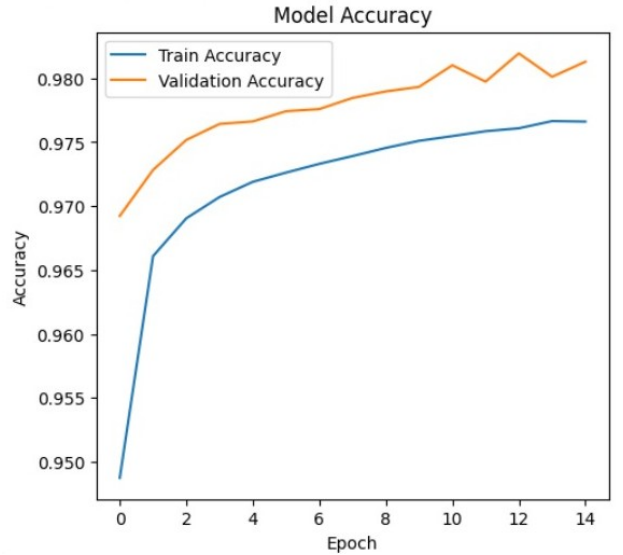


Fig. 3. Model accuracy across all epochs.

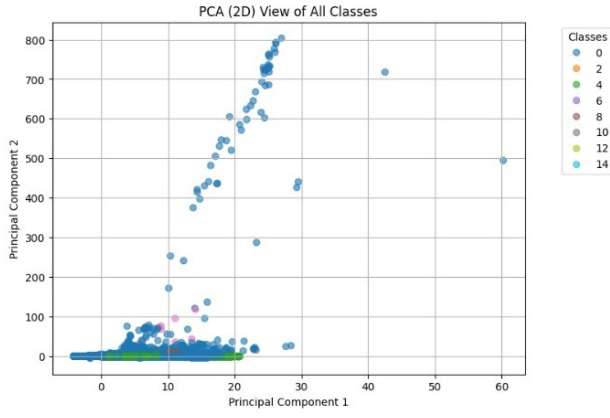


Fig. 4. PCA visualization of network traffic classes.

D. CONFUSION MATRIX ANALYSIS

The confusion matrix (Fig. 5) provides insights into class-wise prediction accuracy. The model accurately identifies most categories such as BENIGN, DDoS, DoS Hulk, and PortScan, with high true positive rates. A small degree of misclassification occurs between similar attack types like DoS Slowloris and DoS GoldenEye, which is expected due to overlapping behaviours in traffic patterns.

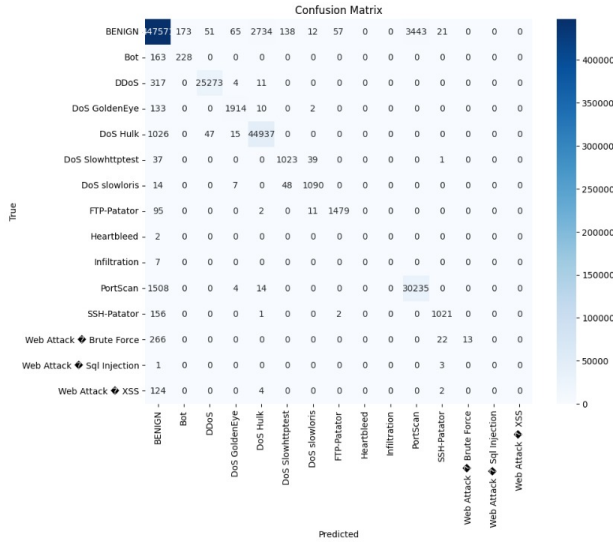


Fig. 5. Confusion matrix for multi-class classification.

This shows the model's robustness in detecting both high-volume and low-volume attacks without significant confusion across categories.

E. COMPARITIVE ANALYSIS

To assess the model's practical effectiveness, we compared its performance with existing models from prior literature.

The proposed system clearly outperforms earlier models in both detection accuracy and balance across classes, making it well-suited for real-time security environments.

TABLE II
COMPARISON WITH RELATED WORKS

Model	Dataset	Accuracy (%)	F1 Score (%)
CNN-GRU [10]	CICIDS2017	96.4	95.8
LSTM-SVM [8]	NSL-KDD	94.2	93.1
Proposed CNN-BiLSTM	CICIDS2017	98.2	98.0

F. DISCUSSION

The superior performance of the model is attributed to the combined strengths of:

- **PCA**, which reduced irrelevant and redundant features.
- **CNN-BiLSTM layers**, which captured fine-grained spatial interactions and modeled the temporal nature of traffic behaviour.
- **Regularization layers** like dropout, which prevented overfitting.

The low false alarm rate of 1.1% ensures that the model is deployable in critical systems like smart cities, financial infrastructures, and industrial IoT environments, where detection delay or false alerts can lead to high risk.

V. CONCLUSION

In this paper, a novel hybrid deep learning model combining PCA, CNN, and BiLSTM was proposed for efficient and accurate cyber threat detection using CICIDS2017 dataset. The system was designed to address few limitations of traditional intrusion detection methods by leveraging spatial and temporal patterns present in network traffic.

The integration of PCA significantly reduced the feature space, enhancing model efficiency without sacrificing important information. CNN layers extracted complex local feature relationships, while BiLSTM units effectively captured the temporal behaviour of traffic sequences. The experimental results demonstrated high detection performance with an accuracy of 98.2%, an F1 score of 98.0%, and a low false alarm rate of 1.1%. The model maintained robustness across multiple classes of attacks, including DoS, DDoS, PortScan, and infiltration attempts.

Furthermore, the confusion matrix, learning curves, and PCA visualization validated the model's stability, learning capacity, and classification capabilities. Compared to existing models, the proposed method offered superior accuracy and lower complexity, making it well-suited for deployment in real-time and resource-constrained environments such as industrial IoT, smart infrastructure, and critical national services. In future work, this framework can be extended to handle encrypted traffic, adapt to adversarial attack conditions, or be integrated into edge-based IDS systems for distributed protection.

REFERENCES

- [1] A. S. Mohamed, A. M. Abdelmaksoud, and M. M. Selim, "Artificial intelligence techniques in cybersecurity: Challenges and future directions," *Knowl. Inf. Syst.*, vol. 67, pp. 1803–1841, 2025.

- [2] S. Abdulkareem *et al.*, “Deep hybrid intrusion detection in IoT by using CNN-BiGRU-BiLSTM optimized with metaheuristics,” *Inf. Syst. Inform.*, vol. 45, no. 2, pp. 310–321, 2023.
- [3] A. Khayat, N. Xiong, and R. Doss, “Adaptive intrusion detection in IoT through reinforcement learning and deep feature engineering,” *arXiv:2205.12466*, 2022.
- [4] R. Srinivasan and R. Senthilkumar, “A composite model for detecting threats in IIoT networks,” in *Proc. Int. Conf. Intell. Comput. Smart Commun.*, 2022, pp. 139–151.
- [5] D. Stephan and A. Mohammed, “Intrusion identification in IoT via SAT-Net and DenseNet fusion,” *Inf. Syst. Inform.*, vol. 44, no. 1, pp. 72–85, 2023.
- [6] M. Akif, N. Ahmad, and A. Javaid, “Ensemble learning-based IDS using IoT-23 benchmark,” *arXiv:2502.12382*, 2024.
- [7] J. Zhang, M. Zulkernine, and A. Haque, “Use of Random Forests for intrusion detection in network environments,” *IEEE Trans. Syst., Man, Cybern. C*, vol. 38, no. 5, pp. 649–659, 2008.
- [8] F. Ullah *et al.*, “LSTM neural networks for recognizing abnormal behavioral patterns,” *Future Gener. Comput. Syst.*, vol. 98, pp. 272–281, 2019.
- [9] A. Vinayakumar, K. P. Soman, and P. Poornachandran, “Deep learning for analyzing malicious traffic in networks,” in *Proc. Int. Conf. Adv. Comput., Commun. Inform. (ICACCI)*, 2017, pp. 1–8.
- [10] S. Hou, D. Li, and W. Ren, “CNN-RNN combination model for anomaly detection in networking,” in *Proc. 4th Int. Conf. Cloud Comput. Internet Things*, 2019, pp. 1–5.
- [11] R. Moustafa and J. Slay, “UNSW-NB15: A benchmark dataset for evaluating intrusion detection,” in *Proc. Mil-CIS Conf.*, 2015, pp. 1–6.
- [12] S. Shone *et al.*, “Applying deep learning models for intrusion detection tasks,” *IEEE Trans. Emerg. Top. Comput. Intell.*, vol. 2, no. 1, pp. 41–50, 2018.
- [13] G. D. Paola *et al.*, “RNN-driven approach for detecting unauthorized access,” in *Proc. Int. Conf. Consumer Electron. (ICCE-Berlin)*, 2018, pp. 1–6.
- [14] J. Hu, S. Liu, and Y. Wang, “Network anomaly detection using CNN and attention-based BiLSTM,” *IEEE Access*, vol. 8, pp. 70116–70125, 2020.
- [15] N. Moustafa and J. Slay, “Performance evaluation of NIDS using UNSW-NB15 and KDD99 datasets,” *Inf. Secur. J.*, vol. 25, no. 1–3, pp. 18–31, 2016.
- [16] M. J. Kang and J. W. Kang, “Securing in-vehicle networks through deep learning-based IDS,” *PLoS One*, vol. 11, no. 6, pp. 1–17, 2016.
- [17] C. Yin *et al.*, “Recurrent neural models for anomaly detection in cyberspace,” *IEEE Access*, vol. 5, pp. 21954–21961, 2017.
- [18] M. Tavallaei *et al.*, “Critical study of the KDD CUP 99 dataset for intrusion detection,” in *Proc. IEEE Symp. Comput. Intell. Secur. Def. Appl.*, 2009, pp. 1–6.
- [19] Y. LeCun, Y. Bengio, and G. Hinton, “Fundamental developments in deep learning,” *Nature*, vol. 521, no. 7553, pp. 436–444, 2015.
- [20] I. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning*. Cambridge, MA, USA: MIT Press, 2016.